SingleRAN

Interoperability Between E-UTRAN and NG-RAN in Idle Mode Feature Parameter Description

Issue Draft A

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Contents

1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-04-27).

- Technical changes

Change Description	Parameter Change	Base Station Model
Added incompatibility of BTS5929R with interoperability between E-UTRAN and NG-RAN in idle mode. For details, see **4.3.3 Hardware**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- Editorial changes

  Revised descriptions in the document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
NR	FOFD-021209	Inter-RAT Mobility from NG-RAN to E-UTRAN	**4 Interoperability Between E-UTRAN and NG-RAN in Idle Mode**
LTE FDD	LNOFD-151336	Inter-RAT Mobility from E-UTRAN to NG-RAN	**4 Interoperability Between E-UTRAN and NG-RAN in Idle Mode**
LTE TDD	TDLNOFD-151501	Inter-RAT Mobility from E-UTRAN to NG-RAN	**4 Interoperability Between E-UTRAN and NG-RAN in Idle Mode**

2.4 Differences

Differences between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
Cell reselection between E-UTRAN and NG-RAN	None	**4 Interoperability Between E-UTRAN and NG-RAN in Idle Mode**

Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
Cell reselection between E-UTRAN and NG-RAN	None	**4 Interoperability Between E-UTRAN and NG-RAN in Idle Mode**

Differences between NSA and SA

Function Name	Difference	Chapter/Section
Cell reselection between E-UTRAN and NG-RAN	Supported in SA networking as well as NSA and SA hybrid networking, but not in NSA networking	**4 Interoperability Between E-UTRAN and NG-RAN in Idle Mode**

Differences between high frequency bands and low frequency bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Cell reselection between E-UTRAN and NG-RAN	Supported only in low frequency bands	**4 Interoperability Between E-UTRAN and NG-RAN in Idle Mode**

3 Overview

Standalone (SA) networking is gaining in popularity for New Radio (NR) deployment. NR operates in high frequency bands (the C-band or higher), causing limited cell coverage. During initial NR network construction, it is difficult to provide continuous coverage. As a result, the coverage continuity of NR networks is worse than that of LTE networks. **Interoperability between evolved universal terrestrial radio access network (E-UTRAN) and Next Generation Radio Access Network (NG-RAN) uses the LTE network with continuous coverage as the basic coverage to ensure service continuity and selects the appropriate bearer network (NR/LTE) according to services to ensure better user experience.**

Interoperability between E-UTRAN and NG-RAN mainly involves the connected and idle modes of UEs. Therefore, interoperability between E-UTRAN and NG-RAN is split into interoperability for UEs in idle mode and that in connected mode. This document describes interoperability between E-UTRAN and NG-RAN in idle mode. For details about interoperability between E-UTRAN and NG-RAN in connected mode, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*. For details about NSA/SA selection based on user experience, see *NSA-SA Selection Based on User Experience*.

NR cells support interoperability with LTE cells in NSA and SA hybrid networking. The interoperability process is the same as that in SA networking.

NOTE

A UE in inactive mode can be transferred from NG-RAN to E-UTRAN through cell reselection, the procedure of which is the same as the reselection from NG-RAN to E-UTRAN for a UE in idle mode. After the reselection to an LTE cell, the UE is in idle mode.

4 Interoperability Between E-UTRAN and NG-RAN in Idle Mode

4.1 Principles

Interoperability between E-UTRAN and NG-RAN in idle mode mainly involves cell reselection. After camping on a cell, a UE in idle mode measures the signal quality of the camped cell and inter-RAT neighboring cells, and then selects a more suitable inter-RAT cell to camp on based on cell reselection criteria.

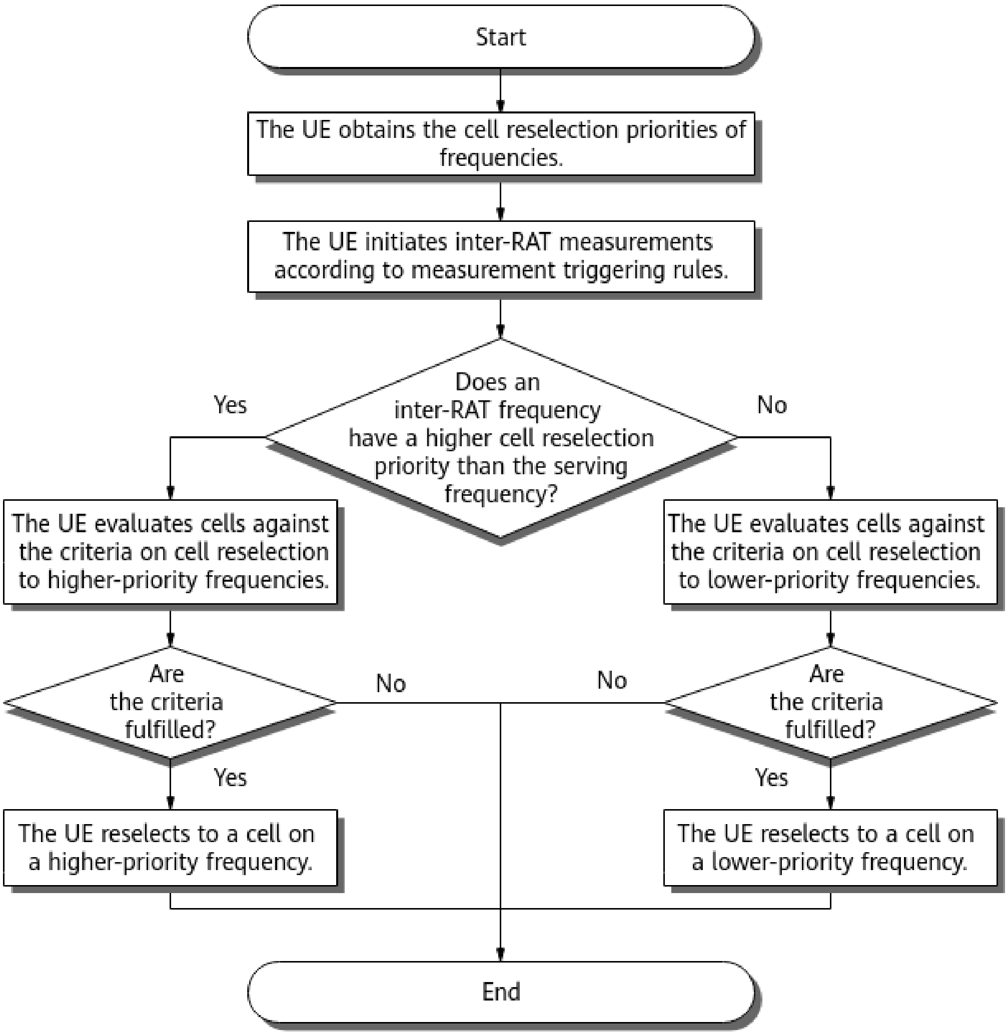
4.1.1 Overall Procedure for Cell Reselection

To enable cell reselection from E-UTRAN to NG-RAN, select the **INTER_RAT_MOBILITY_TO_NR_SW** option of the **CellAlgoExtSwitch**.***HoAllowedSwitch*** parameter.

To enable cell reselection from NG-RAN to E-UTRAN, select the **MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch**.***InterRatServiceMobilitySw*** parameter.

The procedure for cell reselection from E-UTRAN to NG-RAN is similar to that for cell reselection from NG-RAN to E-UTRAN, as shown in **Figure 4-1**.

Figure 4-1 Cell reselection process



In the procedure:

1. The UE checks the cell reselection priorities of frequencies. For details, see **4.1.2 Cell Reselection Priority**.
2. The UE initiates inter-RAT cell measurements according to measurement triggering rules. For details, see **4.1.4 Neighboring Cell Measurement**.
3. The UE reselects to an inter-RAT cell according to the criteria on cell reselection to higher- or lower-priority frequencies.
   - For the criteria on cell reselection to higher-priority frequencies, see **4.1.5 Criteria on Cell Reselection to Higher-Priority Frequencies**.
   - For the criteria on cell reselection to lower-priority frequencies, see **4.1.6 Criteria on Cell Reselection to Lower-Priority Frequencies**.

In the preceding procedure, the base station calculates *Srxlev* and *Squal* of the serving cell and those of inter-RAT neighboring cells. For details, see **4.1.7 Calculation of Srxlev and Squal**.

NOTE

Cells with the same priority do not exist as frequencies of different RATs cannot be configured with the same priority.

4.1.2 Cell Reselection Priority

Cell reselection priorities are classified into common and dedicated priorities.

- Common priorities for cell reselection consist of frequency-level priorities and sub-priorities.
- Dedicated priorities for cell reselection consist of subscriber profile ID (SPID)-specific dedicated priorities, RAT/Frequency Selection Priority (RFSP)-specific dedicated priorities, and operator-specific dedicated priorities.
  - For mobility from E-UTRAN to NG-RAN: SPID-specific dedicated priorities and operator-specific dedicated priorities are supported.
  - For mobility from NG-RAN to E-UTRAN: RFSP dedicated priorities and operator-specific dedicated priorities are supported.

4.1.2.1 Common Priority

A UE compares the reselection priorities of the serving frequency and neighboring frequencies to determine the measurement and reselection objects. The reselection priority is equal to the sum of the cell reselection priority and the cell reselection sub-priority. Frequencies of different RATs must be assigned different priorities.

Table 4-1 lists the parameters that specify the different priority values and sub-priority values.

Table 4-1 Cell reselection priority settings

Scenario	Serving Frequency Priority	Serving Frequency Sub-Priority	Neighboring Frequency Priority Value	Neighboring Frequency Sub-priority Value
Cell reselection from E-UTRAN to NG-RAN	**CellResel**.***CellReselPriority***	N/A	**NrNFreq**.***NrFreqReselPriority***	**NrNFreq**.***NrFreqReselSubPriority***
Cell reselection from NG-RAN to E-UTRAN	**NRCellReselConfig**.***CellReselPriority***	**NRCellReselConfig**.***CellReselSubPriority***	**NRCellEutranNFreq**.***EutranFreqReselPriority***	**NRCellEutranNFreq**.***EutranFreqReselSubPriority***

For cell reselection from E-UTRAN to NG-RAN:

- If the **NrNFreq**.***NrFreqReselPriority*** parameter is set to **255**, no reselection priority is configured or carried for the NR frequency, and system information block 24 (SIB24) does not contain the *cellReselectionPriority-r15* IE of this frequency.

- If the **NrNFreq**.***NrFreqReselSubPriority*** parameter is set to **ZERO**, the sub-priority value is 0, and SIB24 does not contain the *cellReselectionSubPriority-r15* IE.

For cell reselection from NG-RAN to E-UTRAN:

- If the **NRCellEutranNFreq.*****EutranFreqReselPriority*** parameter is set to **255**, no reselection priority is configured or carried for the E-UTRAN frequency, and SIB5 does not contain the *cellReselectionPriority* IE of this frequency.
- If the **NRCellEutranNFreq**.***EutranFreqReselSubPriority*** parameter is set to **ZERO**, the sub-priority value is 0, and SIB5 does not contain the *cellReselectionSubPriority* IE.

Priority Delivery Rules for Cell Reselection from E-UTRAN to NG-RAN

For cell reselection from E-UTRAN to NG-RAN, the priority values and sub-priority values of neighboring NR frequencies are broadcast in SIB24. SIB24 can contain the priority values and sub-priority values of up to eight NR frequencies for one LTE cell. It is recommended that each LTE cell be configured with not more than eight NR frequencies. The eNodeB adheres to the following rules when delivering NR frequencies:

- If no more than eight NR frequencies are configured, all these frequencies are delivered.
- If more than eight NR frequencies are configured, the NR frequencies are sorted in descending order of the sum of the priority value and sub-priority value and the NR frequencies with the highest priorities are selected for delivery. Other frequencies are not delivered.

SIB24 broadcast in LTE cells requires the **Sib24Switch** option of the **CellSiMap**.***SiSwitch*** parameter to be selected, the SIB24 period to be specified using the **CellSiMap**.***Sib24Period*** parameter, and the LTE-to-NR cell reselection timer to be specified using the **CellReselToNr**.***NrCellReselectionTimer*** parameter.

Some legacy LTE UEs cannot identify SIB24-related fields in SIB1 and therefore cannot camp normally on LTE cells. To address this compatibility issue, the **Sib19AndOnwardsSchOptSw** option of the **CellSiMap**.***SibUpdOptSwitch*** parameter has been added to control SIB24 scheduling.

- If this option is selected, SIB24 (together with SIB19 and other SIBs with numbers greater than 19) is scheduled using the *schedulingInfoListExt-r12* IE in SIB1.
- If this option is deselected, SIB24 (together with SIB19 and other SIBs with numbers greater than 19) is scheduled using the *schedulingInfoList* IE in SIB1.

If the **Sib19AndOnwardsSchOptSw** option of the **CellSiMap**.***SibUpdOptSwitch*** parameter is selected for cells serving SA-only or NSA/SA dual-mode 5G UEs that cannot identify the *schedulingInfoListExt-r12* IE, these UEs cannot perform LTE-to-NR cell reselection in idle mode.

Priority Delivery Rules for Cell Reselection from NG-RAN to E-UTRAN

For cell reselection from NG-RAN to E-UTRAN, the priority values and sub-priority values of neighboring LTE frequencies are broadcast in SIB5. SIB5 can

contain the priority values and sub-priority values of up to eight LTE frequencies for an NR cell. The gNodeB adheres to the following rules when delivering LTE frequencies:

- If no more than eight LTE frequencies are configured, all these frequencies are delivered.
- If more than eight LTE frequencies are configured, the LTE frequencies are sorted in descending order of the sum of the priority value and sub-priority value and the LTE frequencies with the highest priorities are selected for delivery. If multiple frequencies are assigned the same priority, frequencies to deliver are randomly selected. Other frequencies are not delivered.

4.1.2.2 Dedicated Priority

4.1.2.2.1 Dedicated Priority for Cell Reselection from E-UTRAN to NG-RAN

Cell reselection from E-UTRAN to NG-RAN supports the following dedicated priorities:

- SPID-specific dedicated priority

  The SPID-specific dedicated priority is delivered to individual UEs to implement differentiated cell reselection policies.
- Operator-specific dedicated priority

  The operator-specific dedicated priority is configured for different operators that share an eNodeB in RAN sharing with common carrier mode.

The SPID-specific dedicated priority takes precedence over the operator-specific dedicated priority. If both priorities exist, the eNodeB preferentially delivers the SPID-specific dedicated priority.

NOTE

- When CA, NSA networking based on EPC, or MLB is also enabled for a cell, the dedicated priority specific to the function is delivered if neither SPID-specific nor operator-specific dedicated priority is configured.
- In roaming scenarios, if no N26 interface is configured between the home LTE core network and the visited NR core network, it is recommended that the **VOICE_NR_HO_POLICY_INDICATION** option of the **NrNFreq**.***AggregationAttribute*** parameter be selected for neighboring NR frequencies that belong to the visited network. In this way, the eNodeB prohibits VoNR-incapable UEs from being transferred to such neighboring NR frequencies through dedicated-priority-based cell reselection.
- For a RedCap UE (the value of the **supportOfRedCap-r17** field in the *UE-NR-Capability* IE is **supported**), the eNodeB filters out an external NR cell when all the following conditions are met: (1) The RedCap UE is not configured with the SPID-specific dedicated priority or operator-specific dedicated priority. (2) The RedCap UE's connection is released and it enters the RRC_IDLE state. (3) The **REDCAP_UE_NR_RESEL_POLICY_SW** option of the **EnodebAlgoExtSwitch**.***NrHandoverAlgoSwitch*** parameter is selected. (4) The RedCap UE supports 1RX or 2RX but the external NR cell does not support 1RX or 2RX (the **NrExternalCell**.***CellBarredRedCap*** parameter is set to a value other than **NOT_BARRED**). If all cells on an NR frequency are filtered out, the NR frequency will not be included in the RRCConnectionRelease message. For details about RedCap UEs, see "RedCap Introduction" in *RedCap* in *5G RAN Feature Documentation*.

SPID-specific Dedicated Priority

SPIDs are policy indexes in the range of 1–256. They are registered by operators for UEs in the database of the home subscriber server (HSS). The eNodeB delivers UE-specific camping and handover policies to UEs based on their SPIDs, ensuring that they camp on or are handed over to appropriate frequencies or RATs according to their subscription information. SPID-specific dedicated priorities are delivered to individual UEs to implement differentiated cell reselection policies. For details about SPID configurations, see *Flexible User Steering* in *eRAN Feature Documentation*.

Before delivering SPID-specific priorities to a UE, the eNodeB uses the SPID-specific frequency list and corresponding priorities configured on the eNodeB to form a list of frequency priorities, and filters the frequencies based on UE capabilities and target PLMNs.

1. The eNodeB filters out NR frequencies not supported by the UE.
2. The eNodeB filters out frequencies based on target PLMNs as follows:
   - If the external cells on a frequency do not belong to any target PLMN, the eNodeB filters out that frequency.
   - If no external cell on a frequency is configured (the PLMN information of external cells is not available), the eNodeB cannot obtain the PLMN information of that frequency and therefore filters out that frequency.

After the filtering, the eNodeB includes the remaining frequencies and their SPID-specific priorities in the *idleModeMobilityControlInfo* (IMMCI) IE in an RRCConnectionRelease message and delivers the message to the UE. This message can contain a maximum of eight neighboring NR frequencies.

According to 3GPP TS 36.304 of Release 15, if the **RatFreqPriorityGroup**.***RatType*** parameter is set to **NR**, the NR Absolute Radio Frequency Channel Numbers (NR-ARFCNs) of the preceding to-be-delivered frequencies (identified by the **RatFreqPriorityGroup**.***DlEarfcn*** parameter) must be already delivered in SIB24. If they have not been delivered in SIB24, the UE will not treat these frequencies as targets for cell reselection.

Operator-specific Dedicated Priority

When the eNodeB is working in RAN sharing with common carrier mode, the operator-specific dedicated priority can be set per PLMN of the serving cell for a neighboring NR frequency. The priority is equal to the sum of **NrNFreqSCellOp**.***CellReselPriority*** and **NrNFreqSCellOp**.***CellReselSubPriority***. The **NrNFreqSCellOp**.***CellReselSubPriority*** parameter must be set to a value other than **ZERO**, and the **NR_FREQ_CELL_RESEL_PRI_OPT_SW** option of the **CellAlgoExtSwitch**.***HoAllowedSwitch*** parameter must be selected. Otherwise, the priority is specified only by the **NrNFreqSCellOp**.***CellReselPriority*** parameter.

The eNodeB finds the operator-specific dedicated priorities of neighboring NR frequencies based on the serving PLMN of a UE, includes these priorities in the IMMCI IE of the RRCConnectionRelease message, and delivers the message to the UE. This message can contain a maximum of eight neighboring NR frequencies. If no dedicated priorities specific to the serving PLMN of the UE are found, operator-specific dedicated priorities are not delivered to the UE.

For a RedCap UE (the value of the **supportOfRedCap-r17** field in the *UE-NR-Capability* IE is **supported**), if the policy for handovers of RedCap UEs to NG-RAN (controlled by the **REDCAP_UE_NR_HO_POLICY_SW** option of the **EnodebAlgoExtSwitch**.***NrHandoverAlgoSwitch*** parameter) is enabled, and the RedCap UE supports 1RX or 2RX but the corresponding external NR cell does not support 1RX or 2RX (the **NrExternalCell**.***CellBarredRedCap*** parameter is set to a value other than **NOT_BARRED**), the eNodeB filters out the NR cell. If all cells on an NR frequency are filtered out, the NR frequency will not be included in the RRCConnectionRelease message. If the *UE-NR-Capability* IE of a RedCap UE does not contain the **supportOfRedCap-r17** field, it is recommended that the **REDCAP_LTE_TO_NR_MOB_SW** option of the **UeCompat**.***WhitelistCtrlExtSwitch1*** parameter be selected for the RedCap UE. In this way, LTE-to-NR mobility for RedCap UEs and the NR handover policy for RedCap UEs can take effect for whitelisted RedCap UEs. For details about LTE-to-NR mobility for RedCap UEs, see *Terminal Awareness Differentiation* in *eRAN Feature Documentation*. For details about RedCap UEs, see "RedCap Introduction" in *RedCap* in *5G RAN Feature Documentation*.

NOTE

For an operator in a cell, if the cell reselection dedicated priority is configured using the **NrNFreqSCellOp** and **EutranNFreqSCellOp** MOs, the sum of the **NrNFreqSCellOp**.***CellReselPriority*** and **NrNFreqSCellOp**.***CellReselSubPriority*** parameter values cannot be the same as the **EutranNFreqSCellOp**.***CellReselDediPri*** parameter value.

Dedicated Priorities Introduced by Other Functions

If one or more of the following functions take effect for the cell as well, and neither SPID-specific nor operator-specific dedicated priorities exist, the dedicated priorities generated by the following functions are delivered and used:

- Primary component carrier (PCC) anchoring for RRC_IDLE UEs in CA

  For CA-capable UEs, when PCC anchoring for RRC_IDLE UEs takes effect, the eNodeB delivers the dedicated frequency priorities generated by this function in the RRC Connection Release message. For details about PCC anchoring for RRC_IDLE UEs, see *Carrier Aggregation*.

- NSA PCC anchoring for RRC_IDLE UEs in NSA networking based on EPC

  For NSA DC-capable UEs, when NSA PCC anchoring for RRC_IDLE UEs takes effect, the eNodeB delivers the dedicated frequency priorities generated by this function in the RRC Connection Release message. For details about NSA PCC anchoring for RRC_IDLE UEs, see *EPC-based NSA Performance Enhancement*.

- MLB by transferring UEs in idle mode

  For UEs selected for MLB, the eNodeB delivers the dedicated frequency priorities generated by this function in the RRC Connection Release message. For details about MLB by transferring UEs in idle mode, see *Intra-RAT Mobility Load Balancing*.

4.1.2.2.2 Dedicated Priority for Cell Reselection from NG-RAN to E-UTRAN

Dedicated priorities for cell reselection from NG-RAN to E-UTRAN consist of the following:

- RFSP-specific dedicated priority

RFSP-specific dedicated priorities are delivered to individual UEs to implement differentiated cell reselection policies.

- Operator-specific dedicated priority

  Operator-specific dedicated priorities are configured for different operators that share a gNodeB in RAN sharing with common carrier mode.

If both of the preceding priorities exist, the gNodeB preferentially delivers RFSP-specific dedicated priorities.

RFSP-specific Dedicated Priority

RFSP indexes range from 1 to 256. They are registered by operators for UEs in the database of the unified data management (UDM). Based on RFSP indexes, the gNodeB delivers UE-specific camping and handover policies to the UEs, ensuring that the UEs camp on or are handed over to appropriate frequencies or RATs according to their subscription information. RFSP-specific dedicated priorities are delivered to individual UEs to implement differentiated cell reselection policies. For details about RFSP configurations, see *Flexible User Steering* in *5G RAN Feature Documentation*.

Before delivering RFSP-specific priorities to a UE, the gNodeB uses the RFSP-specific frequency list and corresponding priorities configured on the gNodeB to form a list of frequency priorities, and filters the frequencies based on UE capabilities and target PLMNs.

1. The gNodeB filters out LTE frequencies not supported by the UE.
2. The gNodeB filters out frequencies based on target PLMNs as follows:
   - If external cells on a frequency are configured and they do not belong to any target PLMN, the gNodeB filters out that frequency.
   - If no external cell on a frequency is configured (the PLMN information of external cells is not available), the gNodeB cannot obtain the PLMN information of that frequency and therefore filters out that frequency.

After the filtering, the gNodeB includes the remaining frequencies and their RFSP-specific dedicated priorities in the *CellReselectionPriorities* IE in an RRCRelease message and delivers the message to the UE. This message can contain a maximum of eight neighboring LTE frequencies.

If the **gNBFreqPriorityGroup**.***RatType*** parameter is set to **EUTRAN**, the EARFCNs of the preceding delivered frequencies (identified by the **gNBFreqPriorityGroup**.***DlEarfcn*** parameter) must be already delivered in SIB5. If they have not been delivered in SIB5, the UE will not treat these frequencies as targets for cell reselection.

Operator-specific Dedicated Priority

When the gNodeB is working in RAN sharing with common carrier mode, operator-specific dedicated priorities can be set per serving PLMN of the shared cell for a neighboring LTE frequency. These priorities are equal to the sum of the **gNBFreqPriorityGroup**.***CellReselPri*** and **gNBFreqPriorityGroup**.***CellReselSubPri*** parameter values. **They are configured for each operator by using the NRCellOpPolicy MO.**

The gNodeB finds the operator-specific dedicated priorities of neighboring LTE frequencies based on the serving PLMN of a UE, includes these priorities in the

CellReselectionPriorities IE of the RRCRelease message, and delivers the message to the UE. This message can contain a maximum of eight neighboring LTE frequencies. **If no dedicated priorities specific to the serving PLMN of the UE are found, operator-specific dedicated priorities are not delivered to the UE.**

Roaming UEs Transferred to Their HPLMNs Through Reselection

To enable the voice-capability-based handover back to the HPLMN, you need to select the **VOICE_CAPB_MOB_TO_HPLMN_SW** option of the **gNBOperator**.***HplmnAlgoSwitch*** parameter on the visited gNodeB for the home network. With this function enabled, the gNodeB identifies VoNR-incapable and VoNR-capable UEs of the home network in roaming scenarios, and performs cell reselection for VoNR-incapable roaming UEs of the home network as follows:

1. The visiting gNodeB sends an RRC Release message to those VoNR-incapable UEs. The message contains the LTE frequencies whose **gNBFreqPriorityGroup**.***RoamingUeHplmnFreqFlag*** is set to **TRUE** in the frequency list bound to the roaming UEs on the gNodeB side.
2. VoNR-incapable roaming UEs are transferred through reselection to the corresponding frequencies based on dedicated cell-reselection priorities.

   For details about how to identify a VoNR-incapable UE on its home network in roaming scenarios and the scenarios where voice-capability-based handover back to the HPLMN is recommended, see the "Inter-RAT Redirection Back to the HPLMN Based on Roaming UE Identification" section in *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

4.1.2.3 Priority Selection

On the E-UTRAN Side

During cell reselection:

- When a UE camping on a suitable cell receives an RRCConnectionRelease message, the UE discards the common cell reselection priorities of NR frequencies, which were obtained from SIB24, if the RRCConnectionRelease message contains dedicated cell reselection priorities.
- When a UE is camping on an acceptable cell, the UE uses the common cell reselection priorities of NR frequencies, which were obtained from SIB24. The UE only stores but does not use dedicated cell reselection priorities, if any.

NOTE

Suitable cells are those in which UEs can obtain normal services.

Acceptable cells are those in which UEs can perform emergency call services only.

A UE discards the dedicated priorities obtained from an RRCConnectionRelease message, if any of the following occurs:

- The non-access stratum (NAS) of the UE instructs the access stratum (AS) to perform PLMN selection.
- The UE enters connected mode.
- Timer T320, which specifies the validity time for the dedicated priorities, expires.

The value of T320 is always 180 minutes. This value is delivered together with the dedicated priorities of frequencies to the UE in the RRCConnectionRelease message. For details about this timer, see section 7.3 "Timers" in 3GPP TS 36.331 V17.2.0.

On the NG-RAN Side

During cell reselection:

- When a UE camping on a suitable cell receives an RRCRelease message, the UE discards the common cell reselection priorities of LTE frequencies, which were obtained from SIB5, if the RRCRelease message contains dedicated cell reselection priorities.
- When a UE is camping on an acceptable cell, the UE uses the common cell reselection priorities of LTE frequencies, which were obtained from SIB5. The UE only stores but does not use dedicated cell reselection priorities, if any.

A UE discards the dedicated priorities obtained from an RRCRelease message, if any of the following occurs:

- The NAS of the UE instructs the AS to perform PLMN selection.
- The UE enters connected mode.
- Timer T320, which specifies the validity time for the dedicated priorities, expires.

  The value of T320 is always 120 minutes. This value is delivered together with the dedicated priorities of frequencies to the UE in the RRCRelease message. For details about this timer, see section 7.1.1 "Timers" in 3GPP TS 38.331 V15.7.0.

4.1.3 Blacklisted Cells for Cell Reselection

When sending system information, the serving cell broadcasts the blacklisted E-UTRAN cells to UEs through the *EUTRA-FreqExcludedCellList* IE in SIB5. During cell reselection, UEs do not measure or reselect to the blacklisted cells broadcast in system information. For details about the *EUTRA-FreqExcludedCellList* IE, see section 6.3.1 "System information blocks" in 3GPP TS 38.331 V17.0.0.

Cell Blacklist Configuration

Blacklisted cells can be configured for a cell in two steps: configuring multiple cell blacklists on the base station, and associating a cell blacklist with the cell.

1. Use the **gNBCellBlacklist** MO on the gNodeB to configure a cell blacklist.
   a. Set the **gNBCellBlacklist**.***CellBlacklistId*** parameter to specify a cell blacklist. Note that the **gNBCellBlacklist**.***RatType*** parameter must be set to **EUTRAN** for the cells to which the **gNBCellBlacklist**.***CellBlacklistId*** is mapped.
   b. Set the start physical cell identifier (PCI) and PCI range to specify a blacklisted cell range. Note that a maximum of 16 blacklisted cell ranges can be configured for each cell blacklist. *N* consecutive PCI-specific cells starting from the one with the specified start PCI are blacklisted.

      Specifically:

- The start physical cell ID is specified by the **gNBCellBlackList**.***StartPhysicalCellId*** parameter.
- The physical cell ID range is specified by the **gNBCellBlackList**.***PhysicalCellIdRange*** parameter.
- *N* is a positive integer to which the value of the **gNBCellBlackList**.***PhysicalCellIdRange*** parameter is mapped.

2. Associate a cell blacklist to a cell on the gNodeB.

   Associate a cell blacklist (specified by the **gNBCellBlacklist**.***CellBlacklistId*** parameter) with a cell using the **NRCellEutranNFreq**.***EutranExcludedCellListId*** parameter. This way, the E-UTRAN cell blacklist associated with the cell determines the blacklisted E-UTRAN cells for the local cell.

NOTE

- If the **NRCellEutranNFreq**.***EutranExcludedCellListId*** parameter is not set to **65535**, its value must be one of the values of the **gNBCellBlackList**.***CellBlacklistId*** parameter, and the corresponding **gNBCellBlacklist**.***RatType*** parameter must be set to **EUTRAN**.
- The values of **gNBCellBlacklist**.***RatType*** in all records with the same **gNBCellBlacklist**.***CellBlacklistId*** value must be the same.

Cell Blacklist Delivery Specifications

SIB5 can indicate a maximum of eight frequencies, and a maximum of 16 blacklisted cell ranges can be configured for each frequency. However, due to the limited length of SIB5, the total number of blacklisted E-UTRAN cell ranges that can be carried in SIB5 is 16.

If there are more than 16 blacklisted E-UTRAN cell ranges in total, 16 blacklisted E-UTRAN cell ranges are selected and signaled in SIB5. The selection is based on the following principles:

- Blacklisted cell ranges associated with high-priority frequencies are preferentially selected. If multiple frequencies have the same priority, blacklisted cell ranges on the E-UTRAN frequencies with a smaller **gNBFreqPriorityGroup**.***DlEarfcn*** parameter value are preferentially selected. For details about frequency priorities, see **4.1.2 Cell Reselection Priority**.
- If not all blacklisted cell ranges on a frequency can be selected, the ones with a larger **gNBCellBlackList**.***PhysicalCellIdRange*** parameter value are preferentially selected. If multiple blacklisted cell ranges have the same **gNBCellBlackList**.***PhysicalCellIdRange*** value, the blacklisted cell ranges with a smaller **gNBCellBlackList**.***StartPhysicalCellId*** parameter value are preferentially selected.

4.1.4 Neighboring Cell Measurement

UEs start inter-RAT neighboring cell measurements for cell reselection according to measurement triggering rules, which are identical for cell reselection from NG-RAN to E-UTRAN and cell reselection from E-UTRAN to NG-RAN. The measurements involve only the neighboring frequencies broadcast in system information or those indicated by RRC connection release messages. The measurement triggering rules are as follows:

- If **the sum of the priority value and sub-priority value** for any inter-RAT **neighboring frequency** indicates a priority level higher than **that for the serving frequency of a UE**, the UE starts to measure inter-RAT cells.
- If **the sum of the priority value and sub-priority value** for any inter-RAT **neighboring frequency** indicates a priority level lower than **that for the serving frequency of a UE**, whether the UE starts to measure inter-RAT cells is based on following rules:
  - If *Srxlev* and *Squal* of the serving cell are greater than $S_{nonIntraSearchP}$ and $S_{nonIntraSearchQ}$, respectively, the UE does not measure inter-RAT cells. For details about how to calculate *Srxlev* and *Squal*, see **4.1.7 Calculation of Srxlev and Squal**.
  - The UE starts to measure inter-RAT cells if *Srxlev* of the serving cell is less than or equal to $S_{nonIntraSearchP}$ or *Squal* of the serving cell is less than or equal to $S_{nonIntraSearchQ}$.

The value of $S_{nonIntraSearchP}$ is determined as follows:

- On the E-UTRAN side, it is specified by the **CellResel.*SNonIntraSearch*** parameter.
- On the NG-RAN side, it is specified by the **NRCellReselConfig.*NonIntraFreqMeasRsrpThld*** parameter.

The value of $S_{nonIntraSearchQ}$ is determined as follows:

- On the E-UTRAN side, it is specified by the **CellResel**.***SNonIntraSearchQ*** parameter. The precondition for the **CellResel**.***SNonIntraSearchQ*** parameter to take effect is that the **CellResel**.***QQualMinCfgInd*** parameter has been set to **CFG**.
- On the NG-RAN side, it is specified by the **NRCellReselConfig**.***NonIntraFreqMeasRsrqThld*** parameter. The setting of the **NRCellReselConfig**.***NonIntraFreqMeasRsrqThld*** parameter can be indicated in SIB2 only if the **NON_INTRA_FREQ_MEAS_RSRQ_SW** option of the **NRCellReselConfig**.***SibOptionalIeInd*** parameter is selected.

To ensure that UEs in idle mode can measure neighboring NR cells, time synchronization is recommended for the eNodeB and the gNodeB. If frequency synchronization is used on the eNodeB and the gNodeB, you are advised to set the SSB period (specified by **NrNFreq**.***SsbPeriod*** on the LTE side and **NRDUCell**.***SsbPeriod*** on the NR side) to **5MS**.

4.1.5 Criteria on Cell Reselection to Higher-Priority Frequencies

Cell Reselection from E-UTRAN to NG-RAN

A UE in an LTE cell is transferred to an NR cell through reselection when all of the following conditions are met:

- The UE has been camping on the serving cell for more than 1 second.
- The signal quality of the NR cell measured by the UE meets one of the following conditions:

  - When the **CellResel**.***ThrshServLowQCfgInd*** parameter is set to **CFG**, *Squal* of the NR cell is continuously greater than *threshX-HighQ-r15* throughout the duration specified by the NR cell reselection timer. The timer and *threshX-HighQ-r15* are broadcast in SIB24 and specified by the **CellReselToNr**.***NrCellReselectionTimer*** and **NrNFreq**.***NrFreqHighPriReselThldRsrq*** parameters, respectively.
  - When the **CellResel**.***ThrshServLowQCfgInd*** parameter is set to **NOT_CFG**, *Srxlev* of the NR cell is continuously greater than *threshX-High-r15* throughout the duration specified by the NR cell reselection timer. The timer and *threshX-High-r15* are broadcast in SIB24 and specified by the **CellReselToNr**.***NrCellReselectionTimer*** and **NrNFreq**.***NrFreqHighPriReselThld*** parameters, respectively.

A cell cannot be a suitable cell for a UE if the cell is included in the list of forbidden tracking areas (TAs) for roaming or if the cell does not belong to the registered PLMN (RPLMN) or an equivalent PLMN (EPLMN). In this case, the UE does not consider this cell as a candidate for reselection within 300 seconds.

Cell Reselection from NG-RAN to E-UTRAN

A UE in an NR cell is transferred to an LTE cell through reselection when all of the following conditions are met:

- The UE has been camping on the serving cell for more than 1 second.
- The signal quality of the LTE cell measured by the UE meets one of the following conditions:
  - When the **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig**.***SibOptionalIeInd*** parameter is selected, *Squal* of the LTE cell is continuously greater than *threshX-HighQ* throughout the duration specified by the E-UTRAN cell reselection timer. The timer and *threshX-HighQ* are broadcast in SIB5 and specified by the **NRCellReselConfig**.***EutranCellReselTimer*** and **NRCellEutranNFreq**.***EutranHighPriReselRsrqThld*** parameters, respectively.
  - When the **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig**.***SibOptionalIeInd*** parameter is deselected, *Srxlev* of the LTE cell is continuously greater than *threshX-High* throughout the duration specified by the E-UTRAN cell reselection timer. The timer and *threshX-High* are broadcast in SIB5 and specified by the **NRCellReselConfig**.***EutranCellReselTimer*** and **NRCellEutranNFreq**.***EutranFreqHighPriReselThld*** parameters, respectively.

A cell cannot be a suitable cell for a UE if the cell is included in the list of forbidden TAs for roaming or if the cell does not belong to the RPLMN or an EPLMN. In this case, the UE does not consider this cell as a candidate for reselection within 300 seconds.

4.1.6 Criteria on Cell Reselection to Lower-Priority Frequencies

Cell Reselection from E-UTRAN to NG-RAN

A UE in E-UTRAN reselects to an NR cell when all of the following conditions are met:

- The criteria for cell reselection to higher-priority frequencies are not met.
- The UE has been camping on the serving cell for more than 1 second.
- The signal quality of the NR cell measured by the UE meets one of the following conditions:
  - The **CellResel**.***ThrshServLowQCfgInd*** parameter is set to **CFG**, in which case the eNodeB broadcasts *threshServingLowQ* (specified by the **CellResel**.***ThrshServLowQ*** parameter) in SIB3. Throughout the duration specified by the NR cell reselection timer, *Squal* of the serving cell is continuously lower than *threshServingLowQ* broadcast in SIB3 and *Squal* of the measured neighboring cell is continuously greater than *threshX-LowQ-r15* broadcast in SIB24. The timer is broadcast in SIB24 and specified by the **CellReselToNr**.***NrCellReselectionTimer*** parameter. *threshX-LowQ-r15* is specified by the **NrNFreq**.***NrFreqLowPriReselThldRsrq*** parameter.
  - The **CellResel**.***ThrshServLowQCfgInd*** parameter is set to **NOT_CFG**, in which case the eNodeB does not broadcast *threshServingLowQ* in SIB3. Throughout the duration specified by the NR cell reselection timer, *Srxlev* of the serving cell is continuously lower than *threshServingLow* broadcast in SIB3 and *Srxlev* of the measured neighboring cell is continuously greater than *threshX-Low-r15* broadcast in SIB24. The timer is broadcast in SIB24 and specified by the **CellReselToNr**.***NrCellReselectionTimer*** parameter. *threshServingLow* and *threshX-Low-r15* are specified by the **CellResel**.***ThrshServLow*** and **NrNFreq.*****NrFreqLowPriReselThld*** parameters, respectively.

Cell Reselection from NG-RAN to E-UTRAN

A UE in NG-RAN reselects to an LTE cell when all of the following conditions are met:

- The criteria for cell reselection to higher-priority frequencies are not met.
- The UE has been camping on the serving cell for more than 1 second.
- The signal quality of the LTE cell measured by the UE meets one of the following conditions:
  - The **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig**.***SibOptionalIeInd*** parameter is selected, in which case the gNodeB broadcasts *threshServingLowQ* (specified by the **NRCellReselConfig**.***ServFreqLowPriRsrqReselThd*** parameter) in SIB2. Throughout the duration specified by the E-UTRAN cell reselection timer, *Squal* of the serving cell is continuously lower than *threshServingLowQ* broadcast in SIB2 and *Squal* of the measured neighboring cell is continuously greater than *threshX-LowQ* broadcast in SIB5. The timer is broadcast in SIB5 and specified by the **NRCellReselConfig**.***EutranCellReselTimer*** parameter. *threshX-LowQ* is

specified by the **NRCellEutranNFreq**.***EutranLowPriReselRsrqThld*** parameter.

– The **RSRQ_BASED_RESEL_SW** option of the **NRCellReselConfig**.***SibOptionalIeInd*** parameter is deselected, in which case the gNodeB does not broadcast *threshServingLowQ* in SIB2. Throughout the duration specified by the E-UTRAN cell reselection timer, *Srxlev* of the serving cell is continuously lower than *threshServingLowP* broadcast in SIB2 and *Srxlev* of the measured neighboring cell is continuously greater than *threshX-Low* broadcast in SIB5. The timer is broadcast in SIB5 and specified by the **NRCellReselConfig**.***EutranCellReselTimer*** parameter. *threshServingLowP* and *threshX-Low* are specified by the **NRCellReselConfig**.***ServFreqLowPriRsrpReselThd*** and **NRCellEutranNFreq**.***EutranFreqLowPriReselThld*** parameters, respectively.

4.1.7 Calculation of *Srxlev* and *Squal*

Srxlev indicates the receive signal strength for cell selection, and *Squal* indicates the receive signal quality for cell selection.

Calculation of *Srxlev*

On both the E-UTRAN and NG-RAN sides, *Srxlev* of the serving cell and that of an inter-RAT neighboring cell are calculated as follows: *Srxlev* = *Qrxlevmeas* – (*Qrxlevmin* + *Qrxlevminoffset*) – $P_{compensation}$.

- $P_{compensation} = \max(P_{EMAX} - P_{PowerClass}, 0)$ (dB)
- $P_{PowerClass}$ is the maximum radio frequency (RF) output power of a UE expressed in dBm. For details about $P_{PowerClass}$ on the E-UTRAN side, see section 6.2.2 "UE maximum output power" in 3GPP TS 36.101 V15.5.0. For details about $P_{PowerClass}$ on the NG-RAN side, see section 6.2.1 "UE maximum output power" in 3GPP TS 38.101-1 V15.5.0.

Table 4-2 and **Table 4-3** list the values of the preceding variables.

Table 4-2 Variables involved in the calculation of *Srxlev* for cell reselection from E-UTRAN to NG-RAN

Variable	Serving LTE Cell (in Reselection from E-UTRAN to NG-RAN)	Neighboring NR Cell (in Reselection from E-UTRAN to NG-RAN)
Qrxlevmeas	Measured receive (RX) signal level of the serving LTE cell	Measured RX signal level of the neighboring NR cell
Qrxlevmin	**CellSel**.***QRxLevMin***	**NrNFreq**.***MinRxLevel***
Qrxlevminoffset	N/A	N/A
P_{EMAX}	**CellResel**.***PMax*** [a]	**NrNFreq**.***MaxAllowedTxPower***

Variable	Serving LTE Cell (in Reselection from E-UTRAN to NG-RAN)	Neighboring NR Cell (in Reselection from E-UTRAN to NG-RAN)
a: On the E-UTRAN side, the **CellResel**.***PMax*** parameter determines the maximum UE transmit power only when the **CellResel**.***PMaxCfgInd*** parameter is set to **CFG**. If the value is not **CFG**, the maximum UE transmit power is determined by the UE capability.		

Table 4-3 Variables involved in the calculation of *Srxlev* for cell reselection from NG-RAN to E-UTRAN

Variable	Serving NR Cell (in Reselection from NG-RAN to E-UTRAN)	Neighboring LTE Cell (in Reselection from NG-RAN to E-UTRAN)
Qrxlevmeas	Measured RX signal level of the serving NR cell	Measured RX signal level of the neighboring LTE cell
Qrxlevmin	**NRDUCellSelConfig**.***MinimumRxLevel***	**NRCellEutranNFreq**.***MinimumRxLevel***
Qrxlevminoffset	N/A	**NRCellEutraNRelation**.***MinimumRxLevelOffset***
P_{EMAX}	23 dBm	**NRCellEutranNFreq**.***MaximumTransmitPower***

Calculation of *Squal*

On both the E-UTRAN and NG-RAN sides, *Squal* of the serving cell and that of an inter-RAT neighboring cell are calculated as follows: *Squal* = *Qqualmeas* – (*Qqualmin* + *Qqualminoffset*).

Table 4-4 and **Table 4-5** list the values of the preceding variables.

Table 4-4 Variables involved in the calculation of *Squal* for cell reselection from E-UTRAN to NG-RAN

Variable	Serving LTE Cell (in Reselection from E-UTRAN to NG-RAN)	Neighboring NR Cell (in Reselection from E-UTRAN to NG-RAN)
Qqualmeas	RX signal quality of the serving LTE cell	RX signal quality of the neighboring NR cell
Qqualmin	**CellSel**.***QQualMin***	**NrNFreq**.***MinRxSignalQuality***
Qqualminoffset	N/A	N/A

Table 4-5 Variables involved in the calculation of *Squal* for cell reselection from NG-RAN to E-UTRAN

Variable	Serving NR Cell (in Reselection from NG-RAN to E-UTRAN)	Neighboring LTE Cell (in Reselection from NG-RAN to E-UTRAN)
Qqualmeas	RX signal quality of the serving NR cell	RX signal quality of the neighboring LTE cell
Qqualmin	**NRDUCellSelConfig**.***MinimumRxQlty***	**NRCellEutranNFreq**.***MinimumRxQlty***
Qqualminoffset	N/A	N/A

4.1.8 Multi-Antenna-Port Configuration Indication

Neighboring LTE cells can be configured with a different number of antenna ports: some with one, and others with two or more. When delivering measurement configurations of LTE frequencies, the gNodeB needs to include the number of antenna ports of the neighboring LTE cells in the LTE measurement configuration for better UE measurement. The multi-antenna-port configuration indication function is introduced for this purpose. This function only applies to the NR side.

This function is recommended for scenarios where at least one neighboring cell on an LTE frequency delivered in SIB5 is configured with a single antenna port. With this function enabled, the value of the *presenceAntennaPort1* IE is **FALSE**. If all neighboring cells on the LTE frequencies delivered in SIB5 are configured with two or more antenna ports, this function should be disabled. In this case, the value of the *presenceAntennaPort1* IE is **TRUE**.

This function is controlled by the **IDLE_PRESENCE_ANTENNA_PORT1_SW** option of the **NRCellEutranNFreq**.***FrequencyCtrlSwitch*** parameter.

4.2 Network Analysis

4.2.1 Benefits

Cell reselection between E-UTRAN and NG-RAN enables UEs to camp on cells with favorable signal quality, ensuring UE service experience.

If the **Sib19AndOnwardsSchOptSw** option of the **CellSiMap**.***SibUpdOptSwitch*** parameter is selected for LTE cells serving LTE UEs that cannot read SIB24-related fields in SIB1, the following KPIs of the cells improve:

- **Average User Number**
- **Downlink Traffic Volume**
- **Uplink Traffic Volume**

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

Some legacy LTE UEs cannot identify SIB24-related fields in SIB1 and therefore cannot camp normally on LTE cells.

If the **Sib19AndOnwardsSchOptSw** option of the **CellSiMap**.***SibUpdOptSwitch*** parameter is selected for cells serving SA-only or NSA/SA dual-mode 5G UEs that cannot identify the *schedulingInfoListExt-r12* IE, these UEs cannot perform LTE-to-NR cell reselection in idle mode.

4.3 Requirements

4.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
NR	FOFD-021209	Inter-RAT Mobility from NG-RAN to E-UTRAN	NR0SRATMNE00	Inter-RAT Mobility from NG-RAN to E-UTRAN (NR)	gNodeB	per cell
LTE FDD	LNOFD-151336	Inter-RAT Mobility from E-UTRAN to NG-RAN	LT1SIRMENG00	Inter-RAT Mobility from E-UTRAN to NG-RAN (LTE FDD)	eNodeB	per cell
LTE TDD	TDLNOFD-151501	Inter-RAT Mobility from E-UTRAN to NG-RAN	LT4SENGMBTDD	Inter-RAT Mobility from E-UTRAN to NG-RAN (LTE TDD)	eNodeB	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	System information broadcast	None	*Idle Mode Management* in *eRAN Feature Documentation*	As new scheduling information is used for SIB19 and SIBs with numbers greater than 19, there will be more scheduling information entries.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all LTE-capable main control boards and BBPs support this function.
- For NR, all NR-capable main control boards and BBPs support this function.

RF Modules

None

4.3.4 Others

- UE requirements

  UEs must comply with 3GPP Release 15 or later.
- Core network requirements

  The core network must comply with 3GPP Release 15 or later.

  When the SIB24 switch is turned on on the LTE side (by selecting the **Sib24Switch** option of the **CellSiMap**.***SiSwitch*** parameter) and UEs in LTE cells reselect to NR cells, the core network must support UE access to SA networks, to prevent UE access failures due to rejection by the core network.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Cell Reselection from E-UTRAN to NG-RAN

Table 4-6 and **Table 4-7** describe the parameters used for activation and optimization, respectively, of cell reselection from E-UTRAN to NG-RAN. These parameters are configured on the eNodeB side. This section does not describe parameters related to cell establishment.

Table 4-6 Parameters used for activation of cell reselection from E-UTRAN to NG-RAN

Parameter Name	Parameter ID	Setting Notes
Handover Allowed Switch	**CellAlgoExtSwitch**.***HoAllowedSwitch***	Select the **INTER_RAT_MOBILITY_TO_NR_SW** option.
Cell System Information Switch	**CellSiMap**.***SiSwitch***	Select the **Sib24Switch** option.

Parameter Name	Parameter ID	Setting Notes
Downlink ARFCN	**NrMfbiFreq**.***DlArfcn***	According to 3GPP TS 38.104, if a frequency belongs to multiple NR frequency bands, the frequency bands must be configured for this frequency on the eNodeB side. Take a neighboring NR frequency whose NR-ARFCN is in the range of 422000–434000 as an example. This frequency belongs to bands n1 and n66. The **NrMfbiFreq** MO on the eNodeB side must be set based on the NR network plan, with this NR frequency configured in band n1 or n66. If a frequency does not belong to multiple NR frequency bands, the NR MFBI parameters do not need to be set.
Frequency Band	**NrMfbiFreq**.***FrequencyBand***	

Table 4-7 Parameters used for optimization of cell reselection from E-UTRAN to NG-RAN

Parameter Name	Parameter ID	Setting Notes
SIB24 period	**CellSiMap**.***Sib24Period***	Set this parameter based on the network plan.
NR Frequency High Priority Reselection Thld	**NrNFreq**.***NrFreqHighPriReselThld***	Set this parameter based on the network plan. A smaller value of this parameter leads to a higher probability of inter-RAT cell reselection. A larger value leads to a lower probability.
NR Frequency Low Priority Reselection Thld	**NrNFreq**.***NrFreqLowPriReselThld***	Set this parameter based on the network plan. A smaller value of this parameter leads to a higher probability of inter-RAT cell reselection. A larger value leads to a lower probability.

Parameter Name	Parameter ID	Setting Notes
Minimum RX Level	**NrNFreq**.***MinRxLevel***	A larger value of this parameter leads to a lower probability that cells on the frequency meet criteria S and become suitable cells. A smaller value leads to the opposite effect. Set this parameter to an appropriate value so that the selected cells can provide signals that meet the quality requirement of basic services.
NR Frequency Reselection Priority	**NrNFreq**.***NrFreqReselPriority***	The greater the sum of the priority value and sub-priority value for the NR frequency, the higher the priority level. This sum value cannot be the same as that for any LTE, UMTS, or GSM frequency. The priority value and sub-priority value for the NR frequency are set separately. The sub-priority value can be normally delivered even if the priority value is set to 255.
NR Frequency Reselection Sub-Priority	**NrNFreq**.***NrFreqReselSubPriority***	The greater the sum of the priority value and sub-priority value for the NR frequency, the higher the priority level. This sum value cannot be the same as that for any LTE, UMTS, or GSM frequency. The priority value and sub-priority value for the NR frequency are set separately. The sub-priority value can be normally delivered even if the priority value is set to 255.
Maximum Allowed Transmit Power	**NrNFreq**.***MaxAllowedTxPower***	The default value is recommended for function activation. A smaller value of this parameter leads to lower signal quality required for selecting cells on the frequency and a lower access success rate of the cells. A larger value leads to the opposite effect.

Parameter Name	Parameter ID	Setting Notes
SSB Subcarrier Spacing	**NrNFreq**.***SubcarrierSpacing***	● Set this parameter to **15KHZ** for NR FDD cells. ● Set this parameter to **30KHZ** for NR TDD cells.
Cell reselection priority	**CellResel**.***CellReselPriority***	A larger value of this parameter leads to a lower probability of reselection to other frequencies. A smaller value of this parameter leads to a higher probability.
Threshold for non-intra freq measurements configure indicator	**CellResel**.***SNonIntraSearchCfgInd***	Set this parameter to **CFG**.
Threshold for non-intra frequency measurements	**CellResel**.***SNonIntraSearch***	The default value is recommended for function activation. A smaller value of this parameter leads to a lower probability of inter-RAT cell reselection. A larger value leads to a higher probability.
Serving frequency lower priority threshold	**CellResel**.***ThrshServLow***	This parameter is shared with other features and has been configured. No additional configuration is required.
Minimum required RX level	**CellSel**.***QRxLevMin***	This parameter is shared with other features and has been configured. No additional configuration is required.
Ue max power allowed configure indicator	**CellResel**.***PMaxCfgInd***	This parameter is shared with other features and has been configured. No additional configuration is required.
Max transmit power allowed	**CellResel**.***PMax***	This parameter is shared with other features and has been configured. No additional configuration is required.

Parameter Name	Parameter ID	Setting Notes
NR Cell Reselection Timer	**CellReselToNr**.***NrCellReselectionTimer***	The default value is recommended for function activation. A smaller value of this parameter leads to a higher probability of inter-RAT cell reselection. A larger value leads to a lower probability. If the value of this parameter is too large or small, the access success rate is affected.
Cell Reselection Priority Config Indicator	**NrNFreqSCellOp**.***CellReselPriorityConfigInd***	In RAN sharing with common carrier scenarios, set this parameter to **CFG** if an operator-specific cell reselection priority needs to be set.
Cell Reselection Priority	**NrNFreqSCellOp**.***CellReselPriority***	Set this parameter based on the network plan.
Cell Reselection Sub-Priority	**NrNFreqSCellOp**.***CellReselSubPriority***	Set this parameter based on the network plan.
Handover Allowed Switch	**CellAlgoExtSwitch**.***HoAllowedSwitch***	The **NR_FREQ_CELL_RESEL_PRI_OPT_SW** option must be selected to make **NrNFreqSCellOp**.***CellReselSubPriority*** take effect.
RSRQ Threshold for non-intra frequency measurements	**CellResel**.***SNonIntraSearchQ***	The default value is recommended for function activation. A smaller value of this parameter leads to a lower probability of inter-RAT cell reselection. A larger value leads to a higher probability.
Serving frequency lower priority RSRQ threshold configure indicator	**CellResel**.***ThrshServLowQCfgInd***	Set this parameter to **CFG** for RSRQ-based cell reselection.
Serving frequency lower priority RSRQ threshold	**CellResel**.***ThrshServLowQ***	The default value is recommended.
NR Frequency High Priority RSRQ Resel Thld	**NrNFreq**.***NrFreqHighPriReselThldRsrq***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
NR Frequency Low Priority RSRQ Resel Thld	**NrNFreq**.***NrFreqLowPriReselThldRsrq***	Set this parameter based on the network plan.
Minimum required RX quality level	**CellSel**.***QQualMin***	Set this parameter to its recommended value.
Minimum RX Signal Quality	**NrNFreq**.***MinRxSignalQuality***	The default value is recommended.
SIB Update Optimization Switch	**CellSiMap**.***SibUpdOptSwitch***	To use the *schedulingInfoList-Ext-r12* IE for scheduling of SIB19 and SIBs with numbers greater than 19, including SIB24, select the **Sib19AndOnwardsSchOptSw** option of this parameter.
Handover to NR Algo Switch	**EnodebAlgoExtSwitch**.***NrHandoverAlgoSwitch***	When not all NR cells support RedCap, it is recommended that the **REDCAP_UE_NR_HO_POLICY_SW** option be selected to prevent RedCap UEs from reselecting NR cells that do not support RedCap based on the dedicated cell reselection priority.
Cell Barred for RedCap	**NrExternalCell**.***CellBarredRedCap***	• When the **REDCAP_SW** option of the NR parameter **NRDUCellFeatureSw**.***UeCapBasedFunSw*** is selected, it is recommended that the LTE parameter **NrExternalCell**.***CellBarredRedCap*** be set to the same value as that of the NR parameter **NRDUCellRedCap**.***CellBarredRedCap***. • When the **REDCAP_SW** option of the NR parameter **NRDUCellFeatureSw**.***UeCapBasedFunSw*** is deselected, it is recommended that the LTE parameter **NrExternalCell**.***CellBarredRedCap*** be set to **1RX_2RX_BARRED**.

Parameter Name	Parameter ID	Setting Notes
Handover to NR Algo Switch	**EnodebAlgoExtSwitch**.***NrHandoverAlgoSwitch***	To enable the LTE-to-NR reselection policy for RedCap UEs, select the **REDCAP_UE_NR_RESEL_POLICY_SW** option.

Cell reselection from NG-RAN to E-UTRAN

Table 4-8 and **Table 4-9** describe the parameters used for activation and optimization, respectively, of cell reselection from NG-RAN to E-UTRAN. These parameters are configured on the gNodeB side.

Table 4-8 Parameters used for activation of cell reselection from NG-RAN to E-UTRAN

Parameter Name	Parameter ID	Setting Notes
Handover Mode Switch	**NRInterRatHoParam**.***HoModeSwitch***	Select either of the following options: ● EUTRAN_HO_SWITCH ● EUTRAN_REDIRECT_SWITCH
Inter-RAT Service Mobility Switch	**NRCellAlgoSwitch**.***InterRatServiceMobilitySw***	Select the **MOBILITY_TO_EUTRAN_SW** option.

Table 4-9 Parameters used for optimization of cell reselection from NG-RAN to E-UTRAN

Parameter Name	Parameter ID	Setting Notes
E-UTRAN Frequency Resel Priority	**NRCellEutranNFreq**.***EutranFreqReselPriority***	Set this parameter based on the network plan. The sum of the priority value and sub-priority value for the LTE frequency cannot be the same as that for any NR frequency. The priority value and sub-priority value for the LTE frequency are set separately. The sub-priority value can be normally delivered even if the priority value is set to 255.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN Frequency Resel Sub-Priority	**NRCellEutranNFreq**.***EutranFreqReselSub-Priority***	Set this parameter based on the network plan. The sum of the priority value and sub-priority value for the LTE frequency cannot be the same as that for any NR frequency. The priority value and sub-priority value for the LTE frequency are set separately. The sub-priority value can be normally delivered even if the priority value is set to 255.
Minimum RX Level	**NRCellEutranNFreq**.***MinimumRxLevel***	This parameter specifies the minimum required receive level for all cells on the LTE frequency. A larger value of this parameter leads to a lower probability that the cell on the frequency meets criteria S, becomes a suitable cell, and is selected as the target cell. A smaller value leads to the opposite effect. Set this parameter to an appropriate value so that the selected cells can provide signals that meet the quality requirement of basic services.
E-UTRAN Frequency High Priority Resel Threshold	**NRCellEutranNFreq**.***EutranFreqHighPriRe-selThld***	The default value is recommended for function activation. A smaller value of this parameter leads to a higher probability of inter-RAT cell reselection. A larger value leads to a lower probability.
E-UTRAN Frequency Low Priority Resel Threshold	**NRCellEutranNFreq**.***EutranFreqLowPriRe-selThld***	The default value is recommended for function activation. A smaller value of this parameter leads to a higher probability of inter-RAT cell reselection. A larger value leads to a lower probability.
Minimum RX Quality	**NRCellEutranNFreq**.***MinimumRxQlty***	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN High Priority Resel RSRQ Thld	**NRCellEutranNFreq**.***EutranHighPriReselRsrqThld***	The default value is recommended for function activation. A smaller value of this parameter leads to a higher probability of inter-RAT cell reselection. A larger value leads to a lower probability.
E-UTRAN Low Priority Resel RSRQ Thld	**NRCellEutranNFreq**.***EutranLowPriReselRsrqThld***	The default value is recommended for function activation. A smaller value of this parameter leads to a higher probability of inter-RAT cell reselection. A larger value leads to a lower probability.
Minimum RX Level Offset	**NRCellEutraNRelation**.***MinimumRxLevelOffset***	The default value is recommended for function activation. If this parameter is set to 0, SIB5 does not contain the *q-RxLevMinOffsetCell* IE. A larger value of this parameter results in a lower probability that the cell meets criteria S for cell reselection, becomes a suitable cell, and is selected as the target cell. A smaller value of this parameter results in a higher probability.
Minimum RX Level	**NRDUCellSelConfig**.***MinimumRxLevel***	This parameter specifies the minimum required receive level for the serving NR cell.
SIB Optional IE Indicator	**NRDUCellSelConfig**.***SibOptionalIeInd***	Select the **SIB1_RSRQ_SW** option.
Minimum RX Quality	**NRDUCellSelConfig.*****MinimumRxQlty***	The default value is recommended.
Cell Reselection Priority	**NRCellReselConfig**.***CellReselPriority***	Set this parameter based on the network plan.
Cell Reselection Sub-Priority	**NRCellReselConfig**.***CellReselSubPriority***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN Cell Reselection Timer	**NRCellReselConfig**.***EutranCellReselTimer***	The default value is recommended for function activation. If this parameter is set to **255**, UEs in the NR cell do not reselect to LTE cells. A smaller value of this parameter leads to a higher probability of inter-RAT cell reselection. A larger value leads to a lower probability. If the value of this parameter is too large or small, the access success rate is affected.
Non-Intra-Freq Measurement RSRP Threshold	**NRCellReselConfig**.***NonIntraFreqMeasRsrpThld***	Set this parameter based on the network plan. A smaller value of this parameter leads to a lower probability of inter-RAT cell reselection. A larger value leads to a higher probability.
Serving Freq Low Priority RSRP Resel Thld	**NRCellReselConfig**.***ServFreqLowPriRsrpReselThd***	You are advised to set this parameter to its recommended value during function activation. A smaller value of this parameter leads to a lower probability of reselection to other frequencies. A larger value leads to a higher probability.

Parameter Name	Parameter ID	Setting Notes
SIB Optional IE Indicator	**NRCellReselConfig**.***SibOptionalIeInd***	• To enable RSRQ-based cell reselection, select the **RSRQ_BASED_RESEL_SW** option. • To use RSRP-based cell reselection, deselect the **RSRQ_BASED_RESEL_SW** option. In this case, SIB5 does not contain the *q-QualMinOffsetCell* IE. If neither the *q-RxLevMinOffsetCell* IE nor the *q-QualMinOffsetCell* IE is contained in the *EUTRA-FreqNeighCellInfo* IE, then the *EUTRA-FreqNeighCellInfo* IE is not included in SIB5. • To enable SIB2 to carry the RSRQ threshold for triggering non-intra-frequency measurements, select the **NON_INTRA_FREQ_MEAS_RSRQ_SW** option.
Serving Freq Low Priority RSRQ Resel Thld	**NRCellReselConfig**.***ServFreqLowPriRsrqRe-selThd***	You are advised to set this parameter to its recommended value during function activation. A smaller value of this parameter leads to a lower probability of reselection to other frequencies. A larger value leads to a higher probability.
Non-Intra-Freq Measurement RSRQ Threshold	**NRCellReselConfig**.***NonIntraFreqMeasRsrqThld***	Set this parameter based on the network plan. A smaller value of this parameter leads to a lower probability of inter-RAT cell reselection. A larger value leads to a higher probability.
Cell Reselection Priority	**gNBFreqPriorityGroup**.***CellReselPri***	Set this parameter based on the network plan. The sum of the priority value and sub-priority value is used as the operator-specific dedicated priority.

Parameter Name	Parameter ID	Setting Notes
Cell Reselection Sub-Priority	**gNBFreqPriorityGroup**.***CellReselSubPri***	Set this parameter based on the network plan. The sum of the priority value and sub-priority value is used as the operator-specific dedicated priority.
Frequency Index	**gNBFreqPriorityGroup**.***FreqIndex***	Set this parameter based on the network plan.
RAT Type	**gNBFreqPriorityGroup**.***RatType***	Set this parameter to **EUTRAN**.
Downlink EARFCN	**gNBFreqPriorityGroup**.***DlEarfcn***	Set this parameter based on the network plan.
gNodeB Frequency Priority Group ID	**gNBFreqPriorityGroup**.***gNBFreqPriority-GroupId***	Set this parameter based on the network plan.
Operator ID	**gNBRfspConfig**.***OperatorId***	Set this parameter based on the network plan.
gNodeB Frequency Priority Group ID	**gNBRfspConfig**.***gNBFreqPriorityGroupId***	Set this parameter based on the network plan.
Index to RAT/ Frequency Selection Priority	**gNBRfspConfig**.***RfspIndex***	Set this parameter based on the network plan.
RAT Type	**gNBCellBlacklist**.***RatType***	Set this parameter based on the network plan.
E-UTRAN Excluded Cell List ID	**NRCellEutranNFreq**.***EutranExcludedCell-ListId***	Set this parameter based on the network plan.
Frequency Control Switch	**NRCellEutranNFreq**.***FrequencyCtrlSwitch***	If at least one neighboring cell on an LTE frequency delivered in SIB5 is configured with a single antenna port, the LTE measurement configuration needs to include the number of antenna ports of the neighboring LTE cells, to improve the measurement accuracy. The **IDLE_PRESENCE_ANTENNA_PORT1_SW** option can be selected for this purpose.

4.4.1.2 Using MML Commands

Activation Command Examples

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

- Cell reselection from E-UTRAN to NG-RAN

```
//Turning on the system information switch for cell reselection from E-UTRAN to NG-RAN
MOD CELLSIMAP: LocalCellId=0, SiSwitch=Sib24Switch-1,
SibUpdOptSwitch=Sib19AndOnwardsSchOptSw-1;
//Setting the SIB24 period
MOD CELLSIMAP: LocalCellId=0, Sib24Period=RF64;
//Turning on the switch for inter-RAT mobility from E-UTRAN to NG-RAN
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=INTER_RAT_MOBILITY_TO_NR_SW-1;
//Setting cell selection parameters
MOD CELLSEL: LocalCellId=0,QRxLevMin=-64, QQualMin=-18;
//Setting reselection parameters for the serving cell
MOD CELLRESEL: LocalCellId=0, CellReselPriority=5,
ThrshServLowQCfgInd=CFG,ThrshServLowQ=1,SNonIntraSearchCfgInd=CFG,SNonIntraSearch=9,SNonIn
traSearchQ=4;
//Setting NR cell reselection parameters
ADD CELLRESELTONR: LocalCellId=0, NrCellReselectionTimer=1;
//Adding a neighboring NR frequency with its cell reselection priority configured (For NR TDD, set
SubcarrierSpacing to 30KHZ.)
ADD NRNFREQ: LocalCellId=0, DlArfcn=428000, NrFreqHighPriReselThld=6, NrFreqLowPriReselThld=7,
NrFreqReselPriority=7, NrFreqHighPriReselThldRsrq=8, NrFreqLowPriReselThldRsrq=9,
SubcarrierSpacing=15KHZ, MaxAllowedTxPower=23, MinRxLevel=-68, NrFreqReselSubPriority=ZERO,
MinRxSignalQuality=0;
//Setting MFBI parameters for the NR frequency
ADD NRMFBIFREQ: DlArfcn=428000, FrequencyBand=N66;
//(Optional; required only when SPID-specific dedicated priorities are used) Adding an external NR cell
ADD NREXTERNALCELL: Mcc="460", Mnc="00", GnodebId=0, CellId=0, DlArfcn=428000,
UlArfcnConfigInd=NOT_CFG, PhyCellId=1, Tac=1;
//(Optional; required only when SPID-specific dedicated priorities are used) Setting the cell reselection
policy based on SPID-specific dedicated priorities
ADD RATFREQPRIORITYGROUP: RatFreqPriorityGroupId=1, RatType=NR, DlEarfcn=428000, Priority=7;
//(Optional; required only when SPID-specific dedicated priorities are used) Adding an SPID
ADD SPIDCFG: Spid=2, RatFreqPriorityInd=CFG, RatFreqPriorityGroupId=1,
VoIPRatFreqPriorityGroupId=1;
//(Optional; required only when the eNodeB is working in RAN sharing with common carrier mode)
Turning on the NR frequency cell reselection priority optimization switch
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=NR_FREQ_CELL_RESEL_PRI_OPT_SW-1;
//(Optional; required only when the eNodeB is working in RAN sharing with common carrier mode)
Setting operator-specific data of the serving cell for the NR frequency
ADD NRNFREQSCELLOP: LocalCellId=0,
DlArfcn=428000,CnOperatorId=0,CellReselPriorityConfigInd=CFG,CellReselPriority=1,CellReselSubPriorit
y=0DOT_6;
//(Optional) Running the following commands for the NR cells that do not support RedCap when not
all NR cells support RedCap
MOD ENODEBALGOEXTSWITCH: NrHandoverAlgoSwitch=REDCAP_UE_NR_HO_POLICY_SW-1;
MOD NREXTERNALCELL: Mcc="460", Mnc="20", GnodebId=123, CellId=1, DlArfcn=428000,
PhyCellId=1, Tac=1, CellBarredRedCap=1RX_2RX_BARRED;
//(Optional) Running the following command when the LTE-to-NR reselection policy for RedCap UEs
needs to be enabled
MOD ENODEBALGOEXTSWITCH: NrHandoverAlgoSwitch=REDCAP_UE_NR_RESEL_POLICY_SW-1;
```

- Cell reselection from NG-RAN to E-UTRAN

```
//Turning on either the handover switch or the redirection switch
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=EUTRAN_HO_SWITCH-1;
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=EUTRAN_REDIRECT_SWITCH-1;
//Turning on the switch for mobility to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=MOBILITY_TO_EUTRAN_SW-1;
//Adding a neighboring LTE frequency for the NR cell with the cell reselection priority of the
frequency configured
ADD NRCELLEUTRANNFREQ: NrCellId=0, DlEarfcn=1300, EutranFreqHighPriReselThld=8,
EutranFreqLowPriReselThld=15, EutranFreqReselPriority=2, MeasurementBandwidth=MBW15,
MinimumRxLevel=-64, EutranFreqReselSubPriority=0DOT4, EutranHighPriReselRsrqThld=10,
EutranLowPriReselRsrqThld=18, MinimumRxQlty=-18;
//Setting the minimum receive level offset (assuming that the mobile country code is 302, the mobile
network code is 220, and the eNodeB ID is 0)
MOD NRCELLEUTRANRELATION: NrCellId=0, Mcc="302", Mnc="220", EnodebId=0, CellId=0,
MinimumRxLevelOffset=1;
//Setting NR DU cell selection parameters
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MinimumRxLevel=-64, MinimumRxQlty=-17,
SibOptionalIeInd=SIB1_RSRQ_SW-1;
//Adding a cell blacklist for the gNodeB
ADD GNBCELLBLACKLIST: CellBlacklistId=1, StartPhysicalCellId=231, PhysicalCellIdRange=N4,
RatType=EUTRAN;
//Modifying the E-UTRAN excluded cell list ID
MOD NRCELLEUTRANNFREQ: NrCellId=0, DlEarfcn=1300, EutranExcludedCellListId=1;
//Setting cell reselection parameters for the NR cell
MOD NRCELLRESELCONFIG: NrCellId=0, CellReselPriority=7, CellReselSubPriority=0DOT0,
EutranCellReselTimer=1, NonIntraFreqMeasRsrpThld=6, ServFreqLowPriRsrpReselThd=6,
NonIntraFreqMeasRsrqThld=7, ServFreqLowPriRsrqReselThd=7,
SibOptionalIeInd=RSRQ_BASED_RESEL_SW-1&NON_INTRA_FREQ_MEAS_RSRQ_SW-1;
//(Optional; required only when RFSP-specific dedicated priorities are used) Adding a frequency
priority group on the gNodeB side
ADD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=EUTRAN,
DlEarfcn=1300, CellReselPri=1, CellReselSubPri=0DOT2;
//(Optional; required only when RFSP-specific dedicated priorities are used) Adding gNodeB RFSP
configurations
ADD GNBRFSPCONFIG: OperatorId=1, RfspIndex=1, gNBFreqPriorityGroupId=0;
//(Optional; required only when the gNodeB is working in RAN sharing with common carrier mode)
Binding the frequency priority group to the NR cell
ADD NRCELLOPPOLICY: NrCellId=0, OperatorId=0, gNBFreqPriorityGroupId=0;
//(Optional) Selecting the IDLE_PRESENCE_ANTENNA_PORT1_SW option when at least one
neighboring cell on an LTE frequency delivered in SIB5 is configured with a single antenna port and
the LTE measurement configuration needs to include the number of antenna ports of the neighboring
LTE cells to improve the measurement accuracy
MOD NRCELLEUTRANNFREQ: NrCellId=0, DlEarfcn=1000, MeasurementBandwidth=MBW6,
FrequencyCtrlSwitch=IDLE_PRESENCE_ANTENNA_PORT1_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Cell reselection from E-UTRAN to NG-RAN

```
//Turning off the switch for mobility from E-UTRAN to NG-RAN
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=INTER_RAT_MOBILITY_TO_NR_SW-0;
//Turning off the system information switch for cell reselection from E-UTRAN to NG-RAN
MOD CELLSIMAP: LocalCellId=0, SiSwitch=Sib24Switch-0;
```

- Cell reselection from NG-RAN to E-UTRAN

```
//Turning off the switch for mobility from NG-RAN to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=MOBILITY_TO_EUTRAN_SW-0;
//Turning off the switches of handover and redirection for inter-RAT mobility from NG-RAN to E-
UTRAN
MOD NRINTERRATHOPARAM: NrCellId=0,
HoModeSwitch=EUTRAN_HO_SWITCH-0&EUTRAN_REDIRECT_SWITCH-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Cell Reselection from E-UTRAN to NG-RAN

The basic observation procedure for cell reselection from E-UTRAN to NG-RAN is as follows:

1. Enable a UE in idle mode to camp on an LTE cell. After the activation operations on the eNodeB are complete, observe the Uu interface messages in the LTE cell. Check that SIB24 carries the *cellReselectionPriority* IE.
2. Move the UE in idle mode to the LTE cell edge, and set up a service so that the UE switches from idle mode to connected mode. Observe the Uu messages in the NR cell. If an RRCSetupRequest message is found, cell reselection has succeeded.

If the **Sib19AndOnwardsSchOptSw** option of the **CellSiMap**.***SibUpdOptSwitch*** parameter is selected, perform the following operation in addition to the preceding basic ones:

Observe the Uu messages in the LTE cell. If SIB24 is carried by the *schedulingInfoListExt-r12* IE in SIB1, this function has taken effect.

Cell Reselection from NG-RAN to E-UTRAN

The basic observation procedure for cell reselection from NG-RAN to E-UTRAN is as follows:

1. Enable a UE in idle mode to camp on an NR cell. After the activation operations on the gNodeB are complete, observe the Uu messages in the NR cell. Check that SIB5 carries the *cellReselectionPriority* IE.
2. Move the UE in idle mode to the NR cell edge, and set up a service so that the UE switches from idle mode to connected mode. Observe the Uu messages in the LTE cell. If an RRCConnectionRequest message is found, cell reselection has succeeded.

The following describes how to observe whether the blacklisted cells for cell reselection function have taken effect.

After related configurations are completed on the gNodeB, observe SIB5 of the current serving cell according to the following steps:

1. Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. On the displayed page, choose **Trace Type** > **NR** > **Application Layer** > **Uu Interface Trace**.
2. If SIB5 delivered by the gNodeB carries the *EUTRA-FreqExcludedCellList* IE, and cells specified by this IE are consistent with the blacklisted cells specified

by the **NRCellEutranNFreq**.***EutranExcludedCellListId*** parameter, the blacklisted cells for cell reselection function has taken effect.

4.4.3 Network Monitoring

None

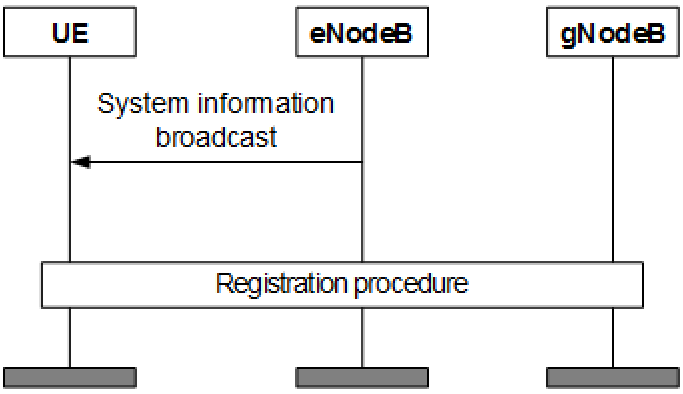
5 Appendix

5.1 Mobility Management for UEs in Idle Mode

Mobility from E-UTRAN to NG-RAN

Figure 5-1 shows the signaling procedure for idle-mode UE mobility from E-UTRAN to NG-RAN. More details can be found in 3GPP TS 36.304 "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) procedures in idle mode".

Figure 5-1 Signaling procedure for cell reselection from E-UTRAN to NG-RAN



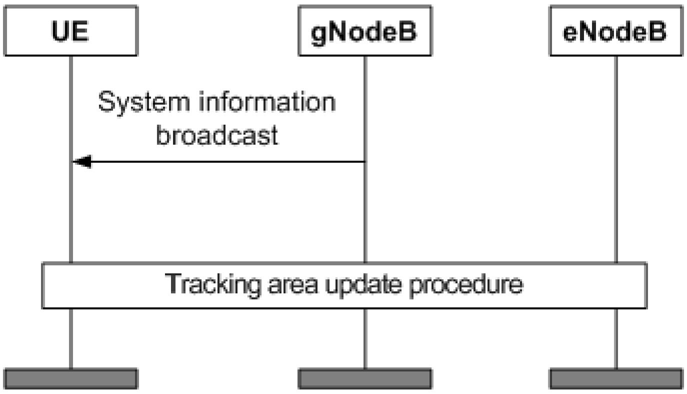
The details are as follows:

1. The UE camps on an LTE cell and receives SIB1, SIB3, and SIB24.
2. The UE determines to reselect to an NR cell according to the criteria for cell reselection to higher- and lower-priority frequencies.
3. The UE initiates an initial registration procedure in the target NR cell.

Mobility from NG-RAN to E-UTRAN

Figure 5-2 shows the signaling procedure for idle-mode UE mobility from NG-RAN to E-UTRAN. More details can be found in 3GPP TS 38.304 "NR; User Equipment (UE) procedures in Idle mode and RRC Inactive state".

Figure 5-2 Signaling procedure for cell reselection from NG-RAN to E-UTRAN



The details are as follows:

1. The UE camps on an NR cell and receives SIB1, SIB2, and SIB5.
2. The UE determines to reselect to an LTE cell according to the criteria for cell reselection to higher- and lower-priority frequencies.
3. The UE initiates an initial tracking area update (TAU) procedure in the target LTE cell.

6 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- *EPC-based NSA Performance Enhancement*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *NSA-SA Selection Based on User Experience*
- *Flexible User Steering* in *eRAN Feature Documentation*
- *Carrier Aggregation* in *eRAN Feature Documentation*
- *Idle Mode Management* in *eRAN Feature Documentation*
- *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*
- *Intra-RAT Mobility Load Balancing* in *eRAN Feature Documentation*
- *Flexible User Steering* in *5G RAN Feature Documentation*
- *Energy Conservation and Emission Reduction* in *5G RAN Feature Documentation*
- *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in *eRAN Feature Documentation*
- *License Control Item Lists* in *5G RAN Feature Documentation*
- *RedCap* in *5G RAN Feature Documentation*
- *Terminal Awareness Differentiation* in *eRAN Feature Documentation*
- 3GPP TS 36.101: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception"
- 3GPP TS 36.304: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) procedures in idle mode"
- 3GPP TS 36.331: "RRC Protocol Specification"
- 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.304: "NR; User Equipment (UE) procedures in Idle mode and RRC Inactive state"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"